

JINUKA SAKITH

Gampaha, Sri Lanka | Jinukasw@gmail.com | +94 71 875 2440 | www.linkedin.com/in/jinuka-waduge-99415735b

PROFILE

First-year Computer Science student at General Sir John Kotelawala Defence University (KDU). Full-stack developer experienced with React, TypeScript, Next.js, and Supabase, with additional depth in embedded systems and control theory (ESP32, PID). Currently preparing for a DevOps internship track (CI/CD, AWS, Docker, Terraform, Linux). Seeking a Software/Full-Stack Development internship to apply and grow these skills in a production environment.

EDUCATION

General Sir John Kotelawala Defence University (KDU), Sri Lanka

B.Sc. in Computer Science | Specializations: AI/ML, Web Development, Embedded Programming

Royal College Colombo, Sri Lanka

GED, A/L | Physical Science Stream – Physics, Combined Mathematics, Chemistry

O/L

ESOFT, Sri Lanka

Diploma in Information Technology

TECHNICAL SKILLS

Web & Full-Stack: React, TypeScript, Next.js 14, Supabase, Tailwind CSS, shadcn/ui, Framer Motion

Systems & Embedded: C++, ESP32, PID Control, Python (PyBullet simulation)

Game Dev / Other: Unity, C#, SFML, CMake, Pygame

DevOps & Tools: Git/GitHub, Linux (Ubuntu Server), Docker, AWS (fundamentals), Terraform (fundamentals), CI/CD concepts

PROJECTS

Bizzles : SME Complaint-to-Proposal Marketplace Platform

Personal Project

Next.js 14 · Supabase · TypeScript · Tailwind CSS v4 · shadcn/ui · Framer Motion

- A two-sided marketplace where local businesses post operational complaints and developers submit funded software proposals to solve them.
- Implementing authentication, database schema, and routing logic end-to-end using Supabase and Next.js middleware.
- Community funded project model for addressing general software issues that add friction to daily business operations.
- Meant to address the lack of platforms that give attention to highly local businesses in the small to medium category and their needs for general purpose, non-bespoke software that eases day-to-day business operations.

3-Axis Servo-Driven Stabilization Platform

Personal Project

ESP32 · MPU6050 IMU · PID Control · Inverse Kinematics · Python · PyBullet

- Built a 3-actuator stabilization platform (Stewart-platform style) using an ESP32 microcontroller and MPU6050 IMU for real-time orientation sensing.
- Developed a PID control loop with inverse kinematics to translate IMU orientation data into precise actuator commands, resolving issues in home-position calibration, absolute IK targeting, and servo pulse range mapping.
- Created a parallel physics simulation in Python (PyBullet) with a Tkinter HUD and a mock ESP32 TCP socket server to test control logic independent of hardware, using complementary/Kalman filter sensor fusion.

CURRENTLY DEVELOPING

- DevOps fundamentals: CI/CD pipelines, AWS, Terraform, and Docker, following a structured self-study roadmap.
- Hands-on Linux administration practice using an Ubuntu Server instance in VirtualBox.
- Covering fundamentals for AI service integration in applications, AI architecture.